



USN

|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Fifth Semester B.Tech (CBCS) Degree Examinations**

December 2024

**EC3203-1- INFORMATION THEORY AND ERROR CONTROL CODING**

(For EC)

Duration: 3 Hours

Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. The information  $I$  in bits contained in a message with probability of occurrence given  $P$  is.  
A)  $I = \log_2 1/P$  B)  $I = \log_2 P$   
C)  $\log_2 1/2P$  D)  $\log_2 1/P^2$
2. For which value(s) of  $p$  is the binary entropy function  $H(p)$  maximized?  
A) 0 B) 0.5  
C) 1 D) 0.25 and 0.75
3. If  $H(s) = 1$  bit/symbol. What is  $H(s^4)$ ?  
A) 1 bit/symbol B) 2 bits/symbol  
C) 4 bits/symbol D) 8 bits/symbol
4. A fair dice is rolled and the outcome is 4. What is the probability of occurrence of this event?  
A)  $\frac{1}{6}$  B)  $\frac{6}{6}$   
C)  $\frac{1}{2}$  D)  $\frac{1}{3}$
5. What is the maximum amount of information content for a source emitting 128 symbols with equal probabilities?  
A) 5 bits/sym B) 7 bits/sym  
C) 64 bits/sym D) 128 bits/sym
6. Using fixed length coding how many bits are needed per symbol to encode 32 different symbols?  
A) 5 bits/symbol B) 10 bits/symbol  
C) 15 bits/symbol D) 25 bits/symbol
7. Which of the following codes are not instantaneous?  
A) {1, 01, 001, 0001} B) {0, 01, 011, 0111}  
C) {0, 10, 110, 111} D) {1, 011, 0101, 001}
8. In dictionary coding the index for the symbol appearing for the first time is  
A) 1 B) 0  
C) Any dummy number D) Highest index in the dictionary
9. What is the dimension of a  $H^T$  matrix of an  $(n, k)$  LBC?  
A)  $k \times n$  B)  $n \times k$   
C)  $(n-k) \times n$  D)  $n \times (n-k)$

10. What is the dimension of a Parity Matrix of an  $(n, k)$  LBC?
- A)  $k \times n$  B)  $n \times k$   
C)  $(n-k) \times k$  D)  $k \times (n-k)$
11. What is the dimension of a Generator matrix of an  $(n, k)$  LBC?
- A)  $k \times n$  B)  $(n-k) \times n$   
C)  $n \times k$  D)  $n \times (n-k)$
12. Consider a Rate  $1/5$  repetition code, what is the error correcting capability of this code?
- A) 1 B) 2  
C) 3 D) 4
13. Consider a Rate  $1/7$  repetition code, what is the error detecting capability of this code?
- A) 3 B) 6  
C) 4 D) 1
14. Number of entries in each row of the standard array for an  $(n, k)$  linear block code is
- A)  $2^n$  B)  $2^k$   
C)  $2^{n-k}$  D)  $2^{n+k}$
15. The first entry of each row of the standard array for an  $(n, k)$  linear block code is called as
- A) Coset Leader B) Coset Member  
C) Parity D) Code Vector
16. Number of valid codewords for a  $(5, 1)$  repetition code is
- A) 5 B) 1  
C) 2 D) 16
17. The maximum number of encoder outputs that can be affected by a single input is called as
- A) Column distance function B) Code length  
C) Code rate D) Constraint length
18. Code rate of a convolution code is  $2/3$ . The number of input lines to the encoder is
- A) 3 B) 2  
C) 1 D) 4
19. The present output of the convolution code is dependent on
- A) Present input only B) Past input only  
C) Present and Past input D) Past Present and Future inputs
20. The length of the output for a  $(2, 1, 3)$  convolution code for the message  $d = (11011)$  is
- A) 12 B) 10  
C) 14 D) 16

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                                                                                                        | Marks | BT* | CO* | PO* |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.       | a) Consider $G = \{0, 1, 2, 3, 4\}$ . Test whether $G$ is a group under addition modulo 5 and multiplication modulo 5.                                                                                 | 8     | L3  | 2   | 1   |
|          | b) Define entropy and state the properties of the entropy.<br>A Zero memory Source with source alphabet, $S = \{S_1, S_2, S_3\}$ , with probability of occurrence, $P = \{0.6, 0.3, 0.1\}$ . Calculate |       |     |     |     |
|          | i) Entropy of the source                                                                                                                                                                               |       |     |     |     |
|          | ii) Entropy of second extension of the given basic source and show that $H(S^2) = 2 H(S)$ .                                                                                                            | 8     | L3  | 1   | 1   |
| 2.       | a) Define field. Investigate whether $R$ , a set of real numbers, forms a Field.                                                                                                                       | 8     | L3  | 2   | 1   |

- b) Given the messages  $x_1, x_2, x_3, x_4, x_5$  and  $x_6$  with respective probabilities of 0.4, 0.2, 0.2, 0.1, 0.08 and 0.02. Construct a binary code by applying Huffman encoding procedure. Determine the efficiency and redundancy of the code so formed.

8 L3 1 1

3. a) Derive an expression for the maximum entropy of a source.  
b) Encode the following input string using LZ algorithm:

8 L3 1 1

'W\_PNW\_NW\_WPN\_P'

8 L3 1 1

## Unit – II

4. a) Consider a systematic (8,4) linear block code whose parity check equations are

$$v_4 = u_1 + u_2 + u_3$$

$$v_5 = u_0 + u_1 + u_2$$

$$v_6 = u_0 + u_1 + u_3$$

$$v_7 = u_0 + u_2 + u_3$$

where  $u_0, u_1, u_2, u_3$  are the message bits

- (i) Write the generator and parity check matrices  
(ii) Draw the encoder diagram

8 L3 3 1

- b) Explain the standard array decoding procedure of an (n, k) linear block code.

8 L2 3 1

5. a) Define Hamming weight, Hamming distance and minimum distance of linear block codes.

Find the generator matrix G and parity check matrix H for a linear block code with minimum Hamming distance 3 and a message block size of 8 bits.

8 L3 3 1

- b) For a (7,4) cyclic code the generator polynomial  $g(x) = 1+x+x^3$ , construct the syndrome calculation circuit. Also obtain the syndrome for the received vector  $r(x) = 1+x^2+x^3+x^4+x^5$ .

8 L3 4 1

6. a) In a linear block code the syndrome is given by,

$$S_1 = r_1 + r_2 + r_3 + r_5$$

$$S_2 = r_1 + r_2 + r_4 + r_6$$

$$S_3 = r_1 + r_3 + r_4 + r_7$$

- i. Find the Parity Check Matrix.  
ii. Draw its encoder circuit

8 L3 3 1

- b) A (15, 5) linear cyclic code has generator polynomial  $g(x) = 1+x+x^2+x^4+x^5+x^8+x^{10}$

Draw the block diagram of an encoder and syndrome calculator for this code.

8 L3 4 1

## Unit – III

7. a) Explain the following with example for a(n,R,M) convolution code

i) Code rate

ii) Constraint length

6 L2 5 1

- b) Consider a (3, 1, 3) with generator polynomials  $g_1 = (1 \ 0 \ 1 \ 1)$ ,  $g_2 = (1 \ 1 \ 1 \ 1)$  and  $g_3 = (1 \ 1 \ 0 \ 1)$ . Draw its encoder diagram. Also find the encoded sequence for the input (1 1 1 0 1 1 0 1) using time-domain approach. Verify the answer using transform-domain approach.

10 L3 5 1

8. a) Explain the procedure of hard decision Viterbi decoding for convolution codes.
- b) Consider the (3,1,2) convolution code with  $g^{(1)}=(110)$ ,  $g^{(2)}=(101)$ ,  $g^{(3)}=(111)$ .
- i) Find the constraint length
  - ii) Find the rate
  - iii) Draw the encoder block diagram
  - iv) Find the generator matrix
  - v) Find the code word for the message sequence (11101) using time domain approach.

|    |    |   |   |
|----|----|---|---|
| 6  | L2 | 5 | 1 |
| 10 | L3 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

\*\*\*\*\*